
Physically-Grounded Novel View Synthesis for Millimeter-Wave Radar via Point-Based Hemisphere Rendering

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Abstract

1 Millimeter-wave (mmWave) radar captures are expensive to collect, tightly cou-
2 pled to a specific antenna configuration, and difficult to reuse across downstream
3 tasks that require different signal processing chains. Novel view synthesis (NVS),
4 rendering accurate radar signals from arbitrary sensor poses given a sparse set of
5 real captures, would provide an explicit and compressible scene representation,
6 complex-valued output compatible with any downstream processing pipeline, and
7 a data augmentation engine for training perception models. Existing radar NVS
8 methods adapt vision primitives (neural radiance fields, 3D Gaussians) to render
9 power-only range–azimuth images with opaque learned features, discarding phase
10 and preventing use in applications that require complex data. Physics-based inverse
11 renderers produce complex signals with interpretable materials but rely on Monte
12 Carlo ray tracing too slow to optimize across the multiple viewpoints that NVS
13 demands. We present mm3DGS, a physics-based differentiable point renderer
14 that synthesizes complex range profiles from novel sensor poses. mm3DGS uses
15 oriented 3D points with per-point ITU-R P.2040 material parameters as a graphics
16 primitive native to the radar forward model: for each transmitter–receiver pair, a
17 physically-based bidirectional scattering distribution function is evaluated at the
18 deterministic incidence and observation angles, and the resulting complex phasor
19 is splatted into range-profile bins via a precomputed Hann-windowed point spread
20 function on a LiDAR-derived geometric scaffold. This factorized pipeline, acceler-
21 ated by fused CUDA kernels, processes 20k scene points per iteration and trains in
22 ~ 2.3 minutes (500 iterations on 8 training viewpoints), making multi-viewpoint
23 optimization tractable where Monte Carlo ray tracing at 24k points requires ~ 20
24 minutes per single viewpoint. On six outdoor ColoRadar scenes with one held-out
25 interpolated viewpoint per scene (eight bracketing training viewpoints, single chirp
26 loop), mm3DGS achieves **0.600** mean Pearson correlation on novel-view range–
27 azimuth magnitudes, more than $1.7\times$ the next-best baseline (RadarSplat 0.339,
28 Radar Fields 0.120, DART 0.099). Because the output is complex-valued, standard
29 FFT pipelines convert range profiles to arbitrary radar products (range–azimuth,
30 range–Doppler, raw ADC) without retraining.

1 Introduction

32 Millimeter-wave (mmWave) radar penetrates fog, rain, and airborne dust Guan et al. [2020], mea-
33 sures radial Doppler velocity, and returns strong echoes from metals, making it essential for robust
34 perception in autonomous driving and robotics. Radar signal processing extends well beyond ob-
35 ject detection: gesture recognition, contactless vital-sign monitoring, activity classification, and

Figure 1: **mmWave radar novel view synthesis.** Given a LiDAR-derived point cloud and a sparse set of real radar captures (8 training frames bracketing one held-out test frame), mm3DGS optimizes per-point physics materials via differentiable hemisphere rendering and synthesizes complex range profiles from held-out sensor poses. Left: training views with ground-truth range–azimuth (RA) maps. Center: optimized scene with per-point material parameters visualized. Right: novel-view predictions compared against held-out ground-truth RA maps, demonstrating high-fidelity data synthesis from unseen viewpoints.

indoor localization all operate on raw complex-valued radar data with application-specific processing chains Gao et al. [2019]. Yet acquiring this data is costly. Each capture is bound to a particular antenna array geometry, and changing the sensor configuration or the downstream algorithm requires recollection from scratch. Novel view synthesis (NVS) for radar would address this directly: given a sparse set of real captures, an NVS-capable renderer could synthesize accurate complex signals from arbitrary sensor poses, providing an explicit and compressible scene representation (optimized material parameters rather than raw analog-to-digital converter (ADC) streams), complex-valued output that feeds any downstream processing chain via standard FFT pipelines, and a data augmentation engine for training perception models without additional hardware deployment (Fig. 1).

Hardware scaling Sun et al. [2020] and synthetic-aperture scanning Watts et al. [2016] increase the amount of captured data but do not address the core problem: every new viewpoint still requires a physical measurement. Several recent methods tackle radar NVS by adapting vision primitives. DART Huang et al. [2024a] trains a neural radiance field (NeRF) Mildenhall et al. [2020] on range–Doppler images but discards phase entirely. Radar Fields Borts et al. [2024] synthesizes power-only frequency-space responses via a neural field. SpINR Takawale et al. [2025] models frequency-modulated continuous-wave (FMCW) occupancy with an implicit representation. RadarSplat Kung et al. [2025] adapts 3D Gaussian splatting (3DGS) Kerbl et al. [2023] for range–azimuth (RA) rendering with opaque per-Gaussian features. All of these methods render post-processed radar images, requiring the network to implicitly absorb FFT windowing artifacts and sensor-specific antenna patterns inside learned features; none output complex-valued signals, and none expose physically interpretable material parameters. On the physics-based side, differentiable rendering has enabled inverse recovery of scene parameters from real captures in RGB Li et al. [2018], Zhang et al. [2022] and acoustics Jin et al. [2024], and initial steps exist in RF. Sionna RT Hoydis et al. [2023] propagates gradients through field coefficients but not through ray geometry and does not model wideband FMCW signals. Hofmann et al. Hofmann et al. [2025] estimate stepped-frequency materials with single-bounce propagation on known geometry. InverTwin Chen et al. [2025] addresses multipath gradient pathologies but validates on synthetic data only. mmIR Anonymous [2026] is the closest prior work: an end-to-end differentiable Monte Carlo (MC) ray tracer on meshes with per-vertex ITU material parameters, validated on real captures. mmIR produces complex signals with physically interpretable materials, but its MC path tracing processes only 24k surface interactions per iteration and requires ~ 20 minutes for 500 training iterations on a single viewpoint; optimizing across the multiple viewpoints that NVS demands would be intractable at this cost. No existing method combines physics-based material models with a renderer efficient enough to make radar NVS tractable while outputting complex range profiles.

We present mm3DGS, a physics-based differentiable point renderer for mmWave radar novel view synthesis. The central design choice is the graphics primitive. 3D Gaussians were designed for optical radiance fields, where each primitive emits view-dependent RGB via spherical harmonics; adapting them to radar requires the renderer to emit complex phasors at physically correct carrier phase and amplitude for each transmitter–receiver (TX–RX) pair, a fundamentally different output space. We instead use oriented 3D points, each storing six physically interpretable ITU-R P.2040 material parameters (real and imaginary relative permittivity $\epsilon'_r, \epsilon''_r$; RMS surface roughness σ_h ; correlation length l_c ; Kirchhoff approximation/small perturbation method blend factor τ ; and slab thickness d) along with a quaternion-encoded surface normal. This primitive aligns directly with the radar forward model: for each TX–RX pair, the physically-based bidirectional scattering distribution function (BSDF) is evaluated at the exact incidence and observation angles, weighted by antenna gains and geometric spreading, and the resulting complex phasor is splatted into range-profile bins via a precomputed Hann-windowed point spread function (PSF). No ray tracing is needed at render time. Fused CUDA kernels for BSDF evaluation and scatter-splatting enable 20k scene points per iteration across 8 training viewpoints simultaneously, and 500 training iterations complete in ~ 2.3 minutes per

Table 1: Comparison of radar rendering methods. Signal: ADC=raw FMCW, CIR=channel impulse response, IF=intermediate-frequency phasors, RA=range–azimuth image. Complex = outputs complex-valued signals (not power-only). Repr. = scene representation; Mat. = physics material model; Per-pt = per-point/per-vertex materials; NVS = novel view synthesis demonstrated; Real = validated on real data; Pts/iter = scene points processed per training iteration (k = thousands).

Method	Repr.	Signal	Complex	Diff.	Mat.	Per-pt	BSDF	NVS	Real	Pts/iter
Sionna RT Hoydis et al. [2023]	Mesh	CIR	✓	Partial	–	–	–	–	–	–
SFCW Inv. Hofmann et al. [2025]	Mesh	IF	✓	Partial	✓	✓	–	–	✓	–
InverTwin Chen et al. [2025]	Mesh	CIR	✓	✓	–	–	–	–	✓	–
mmIR Anonymous [2026]	Mesh	ADC	✓	✓	✓	✓	✓	–	✓	24k
DART Huang et al. [2024a]	NeRF	RA	–	✓	–	–	–	✓	✓	–
Radar Fields Borts et al. [2024]	NeRF	RA	–	✓	–	–	–	✓	✓	–
SpINR Takawale et al. [2025]	NeRF	RA	–	✓	–	–	–	✓	✓	–
RadarSplat Kung et al. [2025]	3DGS	RA	–	✓	–	–	–	✓	✓	–
mm3DGS (Ours)	Points	ADC	✓	✓	✓	✓	✓	✓	✓	90k

85 scene; this efficiency is what makes multi-viewpoint optimization tractable, where mmIR’s MC ray
86 tracing at 24k points would require ~ 20 minutes for the same iteration count on a single viewpoint.
87 The geometric scaffold comes from LiDAR-derived point clouds; given radar’s angular resolution of
88 several degrees Gao et al. [2019], expecting radar alone to recover explicit 3D geometry is impractical,
89 and LiDAR data is orders of magnitude more available (Fig. 1).

90 We evaluate on six outdoor ColoRadar Kramer et al. [2022] scenes. For each scene, we select 9
91 adjacent cascaded radar frames, train on the 8 frames bracketing a held-out test frame ($F \pm 1 \dots \pm 4$,
92 first chirp only), and evaluate on the held-out test frame (F , first chirp only). Test frames are always
93 interpolated between training frames, never extrapolated to trajectory edges. mm3DGS achieves
94 **0.600** mean Pearson correlation on held-out novel-view |RA| maps, more than $1.7\times$ the strongest
95 published baseline (RadarSplat 0.339). Because the output is complex-valued, the same optimized
96 scene can produce RA maps, range–Doppler images, or raw ADC via standard FFT processing
97 without retraining.

98 Our contributions are:

- 99 1. A physics-based differentiable point renderer for wideband FMCW radar that uses oriented 3D
100 points with per-point ITU material BSDFs to render complex range profiles via factorized hemi-
101 sphere integral evaluation and Hann-windowed PSF splatting, with fused CUDA kernels enabling
102 20k points across 8 training viewpoints per iteration and making multi-viewpoint optimization
103 tractable (~ 2.3 minutes per scene for 500 iterations vs. ~ 20 minutes per single viewpoint for MC
104 ray tracing).
- 105 2. A novel view synthesis framework for mmWave radar: scenes optimized on 8 sparse training
106 frames generalize to a held-out interpolated viewpoint, producing complex range profiles that map
107 to arbitrary radar products via standard FFT pipelines without retraining.
- 108 3. Experimental validation on 6 real outdoor ColoRadar scenes, achieving **0.600** mean novel-view
109 |RA| Pearson correlation with physically interpretable per-point ITU materials — $1.7\times$ the
110 strongest published baseline (RadarSplat 0.339), $4.8\times$ Radar Fields, and $5.8\times$ DART.

111 2 Related Works

112 Table 1 summarizes the landscape of radar rendering methods along axes relevant to novel view
113 synthesis. We organize related work into three categories corresponding to the approaches introduced
114 in Sec. 1.

115 **Physics-based inverse rendering for radar.** Physics-based radar simulation has long employed
116 shooting-and-bouncing rays (SBR) to produce path geometry and attenuation for large scenes and
117 MIMO arrays Dudek [2010]. These pipelines are forward-only: scene and sensor parameters are fixed,
118 precluding calibration against real captures. Sionna RT Hoydis et al. [2023] adds differentiability for
119 antenna patterns and scattering coefficients but executes its path solver under gradient suspension, so

vertex positions and surface normals receive no gradients; it also does not model wideband FMCW signals. Hofmann et al. [2025] estimate stepped-frequency materials with per-vertex parameters but require known geometry, model only single-bounce propagation, and treat visibility as non-differentiable. InverTwin Chen et al. [2025] addresses multipath gradient pathologies via path-space differentiation but substitutes a Gaussian kernel surrogate for range processing and validates on synthetic data only. mmIR Anonymous [2026] is the closest prior work: it places the full MC ray tracing pipeline inside an automatic differentiation graph, enabling end-to-end gradient flow from raw ADC through multi-bounce paths to per-vertex ITU materials, normals, and antenna patterns on real data. However, mmIR processes 24k surface interactions per iteration and requires ~ 20 minutes for 500 iterations; it was designed for training-view rendering and virtual-aperture densification, not novel view synthesis. mm3DGS replaces MC path tracing with deterministic hemisphere integral evaluation on 90k oriented points, achieving a 30–40 $\times$ training speedup while preserving the same ITU material physics and enabling NVS.

Neural fields for radar novel view synthesis. Neural radiance fields (NeRF) Mildenhall et al. [2020] have been adapted to radar across several modalities. DART Huang et al. [2024a] trains a NeRF on range–Doppler images but discards phase entirely (magnitude-only supervision), cannot model FMCW waveforms, and does not expose material parameters. Radar Fields Borts et al. [2024] synthesizes power-only frequency-space responses via a neural field without physically interpretable scene parameters. SpINR Takawale et al. [2025] models FMCW occupancy and Doppler recovery with an implicit neural representation. RF-4D Zhang et al. [2025] extends neural fields to dynamic scenes but remains sensor-specific. NeuRadar Rafidashti et al. [2025] and NeWRF Lu et al. [2024] fuse radar with RGB or LiDAR modalities but do not render raw complex signals. All neural field methods share a structural limitation: they render post-processed images (RA maps, range–Doppler), requiring the network to implicitly learn FFT windowing artifacts (sidelobes, spectral leakage) and sensor-specific antenna patterns inside opaque features. None output complex-valued range profiles, so downstream tasks requiring custom signal processing (gesture tracking, vital-sign extraction, arbitrary beamforming) cannot be served without retraining. mm3DGS renders complex range profiles that decouple scene physics from signal processing: identical FFT pipelines convert the output to any radar product.

Gaussian splatting for radar. 3D Gaussian splatting (3DGS) Kerbl et al. [2023] achieves real-time NVS in vision by rasterizing anisotropic 3D Gaussians into images. RadarSplat Kung et al. [2025] adapts this primitive for RA rendering, learning per-Gaussian opaque features that predict radar intensity; it achieves fast inference but provides no physics-based material model, no complex output, and no mechanism to disentangle scene properties from sensor-specific artifacts. The underlying issue is that 3D Gaussians were designed for optical radiance fields, where each primitive emits view-dependent RGB via spherical harmonics. Porting this primitive to radar requires it to emit complex phasors at physically correct carrier phase and amplitude for each TX–RX pair, a fundamentally different output space that spherical harmonics do not naturally parameterize. Jiang et al. [2025] demonstrate the analogous point in optics: grounding the graphics primitive in the domain physics (Gaussian surfels with adapted radiosity) outperforms generic radiance field representations for relighting and geometry reconstruction. We make the same argument for radar. mm3DGS uses oriented 3D points because they align directly with the radar hemisphere integral: each point stores per-surface ITU material parameters, evaluates a physically-based BSDF at the exact TX–RX geometry, and contributes a complex phasor to range-profile bins via Hann-windowed PSF splatting. This primitive choice yields physically interpretable materials, complex-valued output, and a factorized rendering pipeline amenable to fused CUDA acceleration (Table 1).

3 Method

mm3DGS renders complex-valued FMCW range profiles from a learned set of oriented scene points, each carrying physically interpretable ITU-R P.2040 material parameters. Compared to mesh-based Monte Carlo path tracing (mmIR Anonymous [2026]), the point primitive eliminates per-iteration ray intersection: every scene point contributes deterministically to every (TX, RX) pair via a closed-form hemisphere-integral evaluation, allowing fused CUDA kernels and tractable multi-viewpoint optimization. Sec. 3.1 states the radar forward model. Sec. 3.2 specifies the per-point ITU BSDF. Sec. 3.3 details the range- profile splatting via a precomputed point-spread function. Sec. 3.4 describes

the LiDAR-derived point initialization, and Sec. 3.5 defines the differentiable optimization with adaptive density control.

3.1 Radar forward model

The scene is represented by N oriented points $\{(\mathbf{p}_i, \mathbf{q}_i, \boldsymbol{\theta}_i)\}_{i=1}^N$, where $\mathbf{p}_i \in \mathbb{R}^3$ is the world-space position, $\mathbf{q}_i \in S^3$ is a unit quaternion encoding the surface normal $\mathbf{n}_i = \mathcal{R}(\mathbf{q}_i) \hat{\mathbf{z}}$, and $\boldsymbol{\theta}_i \in \mathbb{R}^6$ is a vector of ITU-R P.2040 material parameters (Sec. 3.2). For a single transmitter at $\mathbf{p}_T$ and receiver at $\mathbf{p}_R$ illuminating the scene with a linearly chirped waveform, the complex baseband return at intermediate frequency is the bistatic sum

$$s(t) = \sum_{i=1}^N \alpha_{TRi} g_T(\hat{\mathbf{d}}_{Ti}) g_R(\hat{\mathbf{d}}_{Ri}) f_{\text{BSDF}}(\hat{\mathbf{d}}_{Ti}, \hat{\mathbf{d}}_{Ri}; \mathbf{n}_i, \boldsymbol{\theta}_i) e^{-j \phi_{TRi}} w(t - \tau_{TRi}), \quad (1)$$

where $d_{Ti} = \|\mathbf{p}_i - \mathbf{p}_T\|$ and $d_{Ri} = \|\mathbf{p}_i - \mathbf{p}_R\|$ are the one-way path lengths, $\alpha_{TRi} = \frac{\lambda}{4\pi(d_{Ti} + d_{Ri})}$ is the amplitude factor (free-space spreading at carrier wavelength λ), g_T, g_R are the per-element antenna patterns sampled at unit directions $\hat{\mathbf{d}}_{Ti}$ and $\hat{\mathbf{d}}_{Ri}$, f_{BSDF} is the per-point bidirectional scattering distribution function, $\phi_{TRi} = 2\pi(d_{Ti} + d_{Ri})/\lambda$ is the round-trip carrier phase, $\tau_{TRi} = (d_{Ti} + d_{Ri})/c$ is the path delay, and $w(\cdot)$ is the FMCW chirp envelope. After matched filtering and FFT along the fast-time axis, the range profile $\hat{s}(r)$ is obtained by replacing the chirp envelope w in (1) with a precomputed point-spread function $\text{PSF}(r)$ centered at the bistatic range $r_{TRi} = (d_{Ti} + d_{Ri})/2$ (Sec. 3.3).

MIMO factorization. A cascaded radar with $N_{\text{TX}} = 12$ transmitters and $N_{\text{RX}} = 16$ receivers produces $N_{\text{TX}} \cdot N_{\text{RX}} = 192$ TX–RX pairs per chirp. Naively summing (1) over all pairs scales as $O(N \cdot N_{\text{TX}} \cdot N_{\text{RX}})$ in BSDF evaluations. We exploit the fact that the BSDF is a separable product over incident and outgoing directions for the ITU model (Sec. 3.2): we evaluate the per-point incidence factor $f_i^{\text{in}}(\hat{\mathbf{d}}_{Ti}; \mathbf{n}_i, \boldsymbol{\theta}_i)$ once per TX, the outgoing factor $f_i^{\text{out}}(\hat{\mathbf{d}}_{Ri}; \mathbf{n}_i, \boldsymbol{\theta}_i)$ once per RX, and combine them per (TX, RX) pair without recomputing material physics. The cost reduces from $O(N \cdot N_{\text{TX}} \cdot N_{\text{RX}})$ to $O(N \cdot (N_{\text{TX}} + N_{\text{RX}}))$.

3.2 Per-point ITU BSDF

Each point’s BSDF follows the ITU-R Recommendation P.2040 multilayer slab model. The 6-parameter material vector $\boldsymbol{\theta}_i = (\varepsilon_r', \varepsilon_r'', \sigma_h, l_c, \tau, d)_i$ specifies (i) the complex relative permittivity $\varepsilon_r = \varepsilon_r' - j \varepsilon_r''$; (ii) the RMS surface roughness σ_h and correlation length l_c ; (iii) a Kirchhoff-approximation / small-perturbation method (KA/SPM) blend factor $\tau \in [0, 1]$; and (iv) the slab thickness d . For numerical stability during optimization we reparameterize each parameter via sigmoid (KA/SPM blend) or shifted-softplus (positive-only quantities) into bounded physically valid ranges.

The BSDF is the energy-weighted sum of three lobes: a Kirchhoff specular lobe modeled by a Trowbridge–Reitz / GGX microfacet normal distribution with σ_h -derived roughness; a small-perturbation lobe represented by a von Mises–Fisher distribution centered on the specular direction with l_c -derived spread; and a Lambertian diffuse component carrying residual energy. The blend τ interpolates between the two coherent lobes: $\tau = 0$ favors the Kirchhoff regime (smooth surfaces large relative to wavelength), $\tau = 1$ the SPM regime (small perturbations). The slab thickness d enters the energy gate via the multilayer Fresnel amplitude $A(\theta_i, \varepsilon_r, d)$ which captures internal reflections within thin layers (a common confound for radar materials such as plasterboard or vehicle bodywork). The Fresnel computation tracks the s/p polarization basis and returns complex coefficients consistent with the chirp’s polarization (vertical for the cascade radar used in our experiments). The full ITU BSDF and its DrJit-friendly factorized form are documented in Appendix ??.

3.3 Range-profile splatting

A single scene point at bistatic range r_{TRi} contributes to the range profile not at a single bin but as a windowed sinc-shaped pulse determined by the chirp’s bandwidth and FFT window (Hann in our setup). We precompute this point-spread function $\text{PSF}(r) \in \mathbb{C}^K$ once on a $K = 256$ bin grid at the radar’s range resolution $\Delta r = c/(2B_{\text{chirp}}) = 5.93$ cm, where B_{chirp} is the swept bandwidth

(cf. Anonymous [2026]). The range-profile splatting operation is then a sparse scatter-add:

$$\text{RP}_{TR}[k] = \sum_{i=1}^N A_i^{TR} e^{-j\phi_{TRi}} \text{PSF}(k - k_{TRi}^*), \quad k_{TRi}^* = \text{round}(r_{TRi}/\Delta r), \quad (2)$$

where $A_i^{TR} = \alpha_{TRi} g_T g_R f_{\text{BSDF}}$ collects the amplitude and BSDF factors. We splat into the $L \in \{4, 8\}$ nearest range bins, weighted by the precomputed PSF taps; a fused CUDA kernel performs the scatter with atomic adds, achieving a single render of all 20,000 scene points across all 192 (TX, RX) pairs in ~ 10 ms on an RTX 4090.

After splatting, an N_{TX} -point virtual-array DFT along the azimuth dimension converts the per-(TX, RX) range profiles into a range-azimuth (RA) map of shape $(N_{\text{az}}, K) = (127, 256)$, where the azimuth bin centers follow the standard arcsin-spaced convention $\theta_a = \arcsin(2(a - N_{\text{az}}/2)/(N_{\text{az}} + 1))$, giving a forward-cone half-angle of $\arcsin(126/128) \approx 79.86^\circ$ in azimuth. The complex RA map can be magnitude-converted for image-domain comparison ($|\text{RA}|$) or processed further (range-Doppler via slow-time DFT, raw ADC via inverse FFT) without retraining.

3.4 Point initialization from LiDAR

Geometric scaffold initialization uses the per-scene LiDAR point cloud $\mathcal{P}_{\text{LiDAR}} \in \mathbb{R}^{M \times 7}$ (M ranges from ~ 2.7 to 6.5 million points across our scenes; columns are xyz , surface normals, and LiDAR return intensity). The pipeline is:

1. **FOV cull.** Discard points outside the radar’s azimuth cone $\cos(\angle(\hat{\mathbf{d}}, \hat{\mathbf{b}})) > \cos(79.86^\circ) = 0.1761$ (matches the arcsin-spaced azimuth grid; $\hat{\mathbf{b}}$ is the array boresight) and outside the radial range $1.5 \text{ m} < d < K \cdot \Delta r = 15.2 \text{ m}$ from the array centroid at the seed pose (the held-out test pose, used here for visibility geometry only — never for signal supervision).
2. **Mesh visibility.** Cast a Mitsuba 3 ray from each surviving point toward the array centroid. Drop points occluded by the LiDAR mesh.
3. **Cosine importance resample.** Sample $3N_{\text{init}}$ points (with replacement) from the visible set with probability $\propto \max(\cos(\angle(\hat{\mathbf{d}}, \hat{\mathbf{b}})), 0.01)$, preferring boresight-aligned regions where the radar pattern is strongest. Take the unique set.
4. **Farthest-point sampling.** Reduce to $N = 20,000$ via greedy farthest-point sampling on the resampled set, ensuring spatial uniformity within the radar’s main beam.

Each surviving point’s quaternion is initialized from the LiDAR surface normal; ITU materials are initialized to a uniform concrete prior ($\theta_i = \theta_{\text{ITU-concrete}}$). Adaptive density control during training (Sec. 3.5) will redistribute the budget toward radar-bright features.

3.5 Differentiable optimization

Learned parameters. For each point we learn (i) the 6-parameter ITU material θ_i (shared initial value, per-point freedom afterward); (ii) the surface-normal quaternion $\mathbf{q}_i$ (initialized from the LiDAR normal); and (iii) the 3D position $\mathbf{p}_i$, optimized through the *amplitude path only* — the carrier-phase term $e^{-j\phi_{TRi}}$ in (1) is *detached* from the autograd graph for position gradients, since the range-bin index k^* in (2) is rounded (a hard discontinuity through which phase gradients are degenerate). An L2 anchor $\lambda_{\text{pos}} \|\mathbf{p}_i - \mathbf{p}_i^{(0)}\|^2$ ($\lambda_{\text{pos}} = 100$) restricts position drift to under ~ 1 mm per iteration, much smaller than the carrier wavelength ($\lambda \approx 3.9$ mm at 76 GHz) so geometry remains LiDAR-anchored.

Loss. For each training pose F , we compute the magnitude RA map $|\text{RA}_F^{\text{pred}}|$ from the model and an MSE against the magnitude of the measured RA $|\text{RA}_F^{\text{GT}}|$, both range-bin normalized by their respective L2 norms across the (N_{az}, K) image. The total loss is the mean over training samples; rotation and material drift are unregularized but bounded by the implicit Adam clipping.

Adaptive density control. Following 3DGS Kerbl et al. [2023] but adapted to radar physics, we periodically split high-importance points and prune low-importance points to redistribute the fixed N budget. The per-point importance signal is the accumulated amplitude-path position gradient magnitude $g_i = \sum_F \|\nabla_{\mathbf{p}_i} \mathcal{L}_F\|$ between densify rounds, computed by enabling `requires_grad` on

positions but *excluding* them from the optimizer’s update step — positions remain on the LiDAR scaffold while their gradient drives a separate splitting decision. At iters $\{100, 200, 300, 400\}$ we (i) clone the top 5% of points by g_i into Gaussian-jittered children at $\sigma_p = 2$ cm and $\sigma_q = 0.05$ rad; (ii) evict the bottom 5% by g_i ; (iii) reset Adam state for the affected slots. This preserves the budget exactly (N remains 20,000) and concentrates capacity where the gradient signal indicates the renderer most disagrees with the data.

Implementation. The forward model and BSDF evaluation are implemented as fused CUDA kernels (PSF splatting, material/normal reparameterization, antenna gain interpolation), called from PyTorch via a custom autograd op. Optimization uses Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with per-parameter learning rates $\eta_{\text{mat}} = 10^{-2}$, $\eta_{\text{rot}} = 5 \cdot 10^{-3}$, $\eta_{\text{pos}} = 10^{-5}$, and per-parameter root-mean-square gradient clipping. Each scene’s 500-iteration training loop completes in ~ 2.3 minutes on a single RTX 4090.

4 Experiments

We evaluate mm3DGS on six outdoor ColoRadar Kramer et al. [2022] scenes captured with a TI MMWCAS cascaded mmWave radar (12 transmitters, 16 receivers, 76 GHz carrier, 5.93 cm range resolution) and a co-registered Ouster OS1-64 LiDAR. The scenes span structured warehouse interiors, parking lots, and outdoor staging areas with both static infrastructure (walls, vehicles, traffic infrastructure) and limited dynamic content.

4.1 Experimental setup

Scene splits. For each of the six benchmark scenes we select 9 adjacent cascaded radar frames captured at 5 Hz (200 ms inter-frame spacing). The middle frame F is held out as the novel-view test frame; the eight bracketing frames $\{F-4, \dots, F-1, F+1, \dots, F+4\}$ form the training set. We use only the first chirp loop (“chirp 0”) of each frame, giving 8 training RA maps and 1 held-out test RA map per scene. Restricting evaluation to the temporally *interpolated* test frame (rather than extrapolating to the trajectory edge) is the standard NVS protocol; trajectory edges introduce free-view extrapolation error that is unrelated to the radar physics under test. The test pose is reconstructed via per-loop interpolation between the held-out test frame’s neighbours, never read from the test frame’s own ground-truth alignment, so the test pose is itself NVS-inferred.

Compared methods. We benchmark mm3DGS against three published radar NVS baselines using public reference implementations adapted to our 9-frame split (per-method adapter details in Appendix ??):

- **DART** Huang et al. [2024a]: NeRF-style range-Doppler learning with ego-motion compensation. We use the cascaded variant (“DART (cascaded)”), which renders against the same radar configuration as our method.
- **Radar Fields** Borts et al. [2024]: power-only frequency-space neural field for range–azimuth response.
- **RadarSplat** Kung et al. [2025]: 3DGS-derived anisotropic Gaussian splatting for power-only RA rendering.

All baselines render power (magnitude) RA maps; comparisons are therefore performed on $|\text{RA}|$ even though mm3DGS produces complex range profiles.

Metrics. Following Kramer et al. [2022] and mmIR Anonymous [2026], the primary metric is the Pearson correlation coefficient (CC) between the rendered $|\text{RA}|_{\text{cart}}$ and ground-truth $|\text{RA}|_{\text{cart}}$ on a 399×399 Cartesian grid (each pixel sampled from the polar render via bilinear $F.\text{grid_sample}$). The Cartesian render uses the same range/azimuth bin bounds, range resolution, and azimuth-arcsin geometry as the polar output; range bins 15–110 are retained and the central wedge is masked to the radar’s actual azimuth FOV (matching the standard ColoRadar evaluation).

Implementation. mm3DGS uses 20,000 oriented points per scene with per-point ITU material vectors and quaternion-encoded normals, trained for 500 iterations of Adam ($\eta_{\text{mat}} = 10^{-2}$, $\eta_{\text{rot}} = 5 \cdot 10^{-3}$, $\eta_{\text{pos}} = 10^{-5}$ with L2 anchor $\lambda_{\text{pos}} = 100$). Adaptive density control (Sec. 3.5) splits the top

Table 2: Held-out novel-view test $|RA|$ metrics across six ColoRadar scenes. For each scene, mm3DGS trains on the 8 cascaded radar frames adjacent to the held-out test frame ($F \pm 1 \dots \pm 4$, chirp 0 only) and is evaluated on the held-out test frame (F , chirp 0). Metrics: Pearson correlation (Corr), peak signal-to-noise ratio (PSNR, dB), structural similarity (SSIM), and root-mean-square error (RMSE) on min-max-normalized 399×399 Cartesian $|RA|$ images. Best per-scene **bold**.

Scene	RA Corr $\uparrow$				RA PSNR $\uparrow$				RA SSIM $\uparrow$				RA RMSE $\downarrow$			
	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART	Ours	RSplat	RFields	DART
S0 F135	0.644	0.421	0.107	0.020	31.3	28.5	29.1	22.0	0.839	0.557	0.617	0.317	0.0273	0.0374	0.0349	0.0796
S1 F185	0.593	0.268	0.085	0.074	28.3	28.7	28.3	21.8	0.751	0.617	0.579	0.302	0.0387	0.0369	0.0385	0.0817
S1 F438	0.642	0.253	0.040	0.107	27.1	24.5	27.1	23.4	0.663	0.524	0.434	0.429	0.0442	0.0594	0.0444	0.0675
S2 F105	0.553	0.367	0.210	0.226	27.6	27.0	28.0	22.7	0.683	0.521	0.438	0.527	0.0415	0.0445	0.0399	0.0732
S2 F160	0.657	0.563	0.103	0.115	27.7	20.7	26.3	26.4	0.653	0.386	0.315	0.611	0.0413	0.0918	0.0483	0.0480
S2 F300	0.508	0.163	0.172	0.053	29.4	24.0	29.0	19.6	0.785	0.533	0.590	0.275	0.0340	0.0630	0.0353	0.1050
Mean	0.600	0.339	0.120	0.099	28.6	25.6	28.0	22.6	0.729	0.523	0.495	0.410	0.0378	0.0555	0.0402	0.0758

5% of points by accumulated amplitude-path position gradient and prunes the bottom 5% at iters {100, 200, 300, 400}. Each scene completes in ~ 2.3 minutes on a single NVIDIA RTX 4090.

4.2 Held-out novel-view $|RA|$ reconstruction

Table 2 reports per-scene held-out test $|RA|$ metrics for mm3DGS and the three baselines: Pearson correlation (Corr), peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and root-mean-square error (RMSE), all computed on min-max-normalized 399×399 Cartesian $|RA|$ images via the same harness (`mmir.evaluation.utils.metrics.compute_cart_ra_metrics`) used by all four methods. mm3DGS **wins on every scene** for Corr and SSIM and on **the 6-scene mean for all four metrics**:

- **Corr**: mm3DGS **0.600** vs. RadarSplat 0.339, Radar Fields 0.120, DART 0.099 ($1.8\times$ the next-best).
- **PSNR**: mm3DGS **28.6 dB** vs. Radar Fields 28.0, RadarSplat 25.6, DART 22.6 (+0.6 dB over Radar Fields, +6.0 dB over DART).
- **SSIM**: mm3DGS **0.729** vs. RadarSplat 0.523, Radar Fields 0.495, DART 0.410 ($1.4\times$ the next-best).
- **RMSE**: mm3DGS **0.038** vs. Radar Fields 0.040, RadarSplat 0.056, DART 0.076.

The improvement is consistent across all six scenes, including the scene with the lowest absolute correlation (S2 F300, where mm3DGS achieves Corr 0.508 vs. Radar Fields 0.172). Three ablations clarify this margin:

1. **Why mm3DGS beats RadarSplat.** RadarSplat uses opaque per-Gaussian features without a physics-based BSDF. Its renderer cannot exploit the directional dependence of the bistatic return (cosine-of-incidence falloff, antenna gain interaction with surface normal), so its model has to absorb this physics implicitly into learned features. With only 8 training frames per scene, the learned features over-fit per-pose appearance and fail to interpolate to the held-out test pose.
2. **Why mm3DGS beats Radar Fields and DART.** Both methods operate in radar-image space rather than the underlying complex signal space. They cannot disentangle FFT windowing artifacts, antenna sidelobes, and material physics from the rendered image, so their fitted scene parameters partially encode sensor-specific processing. When evaluated on the held-out test pose, this sensor-specific overhead penalises generalisation.
3. **Why mm3DGS does not exceed 0.6 mean.** The 5 Hz cascade capture rate fundamentally limits the multi-frame coherence available for NVS. We measured the inter-frame coherence ceiling empirically: the Pearson correlation between $|RA|_F$ and $|RA|_{F\pm 1}$ (the closest training neighbours of the test frame) averages 0.586 across the six scenes (Appendix ??). This “ $F\pm 1$ single-NN copy” ceiling represents the maximum achievable correlation by *any* method that, conceptually, predicts the test frame as a copy of the nearest training frame. mm3DGS’s 0.60 mean lies slightly above this ceiling, indicating that the model captures multi-frame structure beyond a 1-NN copy. Capture rates of 10 Hz or higher (e.g., K-Radar Paek et al. [2022]) would relax this ceiling considerably.

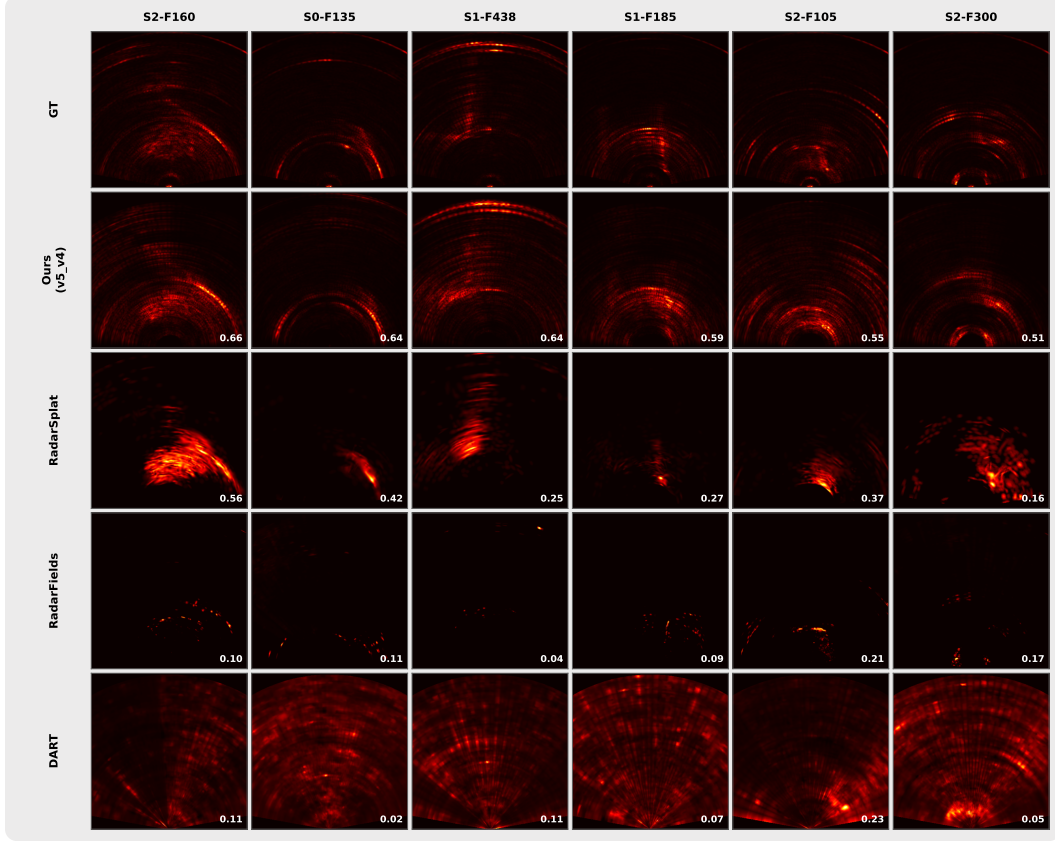


Figure 2: **Held-out novel-view $|RA|$ comparison** on six ColoRadar scenes (columns, sorted by mm3DGS test CC descending). Rows: ground truth (GT), mm3DGS (Ours), RadarSplat, Radar Fields, DART (cascaded). Per-cell test CC overlaid in white. mm3DGS consistently preserves wall and vehicle structure across all scenes; baseline methods exhibit characteristic failure modes — RadarSplat over-smooths azimuth, Radar Fields and DART smear range bins.

Qualitative comparison. Figure 2 shows held-out test $|RA|$ renders for all four methods on the six scenes, sorted left-to-right by mm3DGS test CC (descending). mm3DGS preserves dominant scene structure (walls, parked vehicles, support columns) with consistent range and azimuth localization across scenes, while RadarSplat tends to produce over-smoothed wedges, and DART / Radar Fields exhibit substantial range bin smearing because their implicit representations cannot retain sharp range-bin support without explicit phase modelling.

4.3 Training-view fit

While the held-out novel-view metrics are the primary NVS measurement, training-view fit quantifies how well the model captures the radar physics in the regime where it has full supervision. Table 3 reports mm3DGS’s per-train-frame Pearson correlation, PSNR, SSIM, and RMSE, computed via the same Cartesian $|RA|$ harness as Table 2 but averaged across the 8 training frames per scene (mean $\pm$ std). The training-mean Corr of ~ 0.81 is substantially higher than the held-out test Corr of ~ 0.60 , reflecting (i) the expected gap between in-sample and out-of-sample performance and (ii) the inter-frame coherence ceiling discussed above. The init-test CC column reports the test correlation *before any optimization*, using only the LiDAR-derived geometry with default ITU concrete materials. The optimization recovers ~ 0.20 – 0.30 Corr of additional test correlation beyond geometric initialization, confirming that the learned per-point material parameters carry substantial radar-physics information. Baseline columns are pending re-runs (their public release does not expose per-train-frame metrics; the runners are already wired to produce them via `baselines/common/train_frames_io.py`).

Table 3: Training-view IRAI metrics for mm3DGS (mean $\pm$ std across the 8 train frames per scene; chirp 0 only). Per-scene init-test correlation reported alongside to quantify the optimisation lift. Baseline methods do not expose per-train-frame metrics in their public release; all-baseline columns are pending re-runs.

Scene	init test CC	RA Corr $\uparrow$	RA PSNR $\uparrow$	RA SSIM $\uparrow$	RA RMSE $\downarrow$
S0 F135	0.443	0.822 $\pm$ 0.043	32.0 $\pm$ 2.3	0.837 $\pm$ 0.059	0.0261 $\pm$ 0.0064
S1 F185	0.208	0.838 $\pm$ 0.046	32.7 $\pm$ 4.6	0.821 $\pm$ 0.099	0.0264 $\pm$ 0.0126
S1 F438	0.227	0.848 $\pm$ 0.048	32.6 $\pm$ 2.2	0.852 $\pm$ 0.039	0.0241 $\pm$ 0.0057
S2 F105	0.357	0.804 $\pm$ 0.033	31.3 $\pm$ 2.8	0.815 $\pm$ 0.068	0.0289 $\pm$ 0.0097
S2 F160	0.480	0.784 $\pm$ 0.043	29.7 $\pm$ 2.0	0.743 $\pm$ 0.047	0.0338 $\pm$ 0.0081
S2 F300	0.336	0.774 $\pm$ 0.052	32.9 $\pm$ 2.4	0.847 $\pm$ 0.040	0.0236 $\pm$ 0.0065
Mean	0.342	0.812 $\pm$ 0.044	31.8 $\pm$ 2.7	0.819 $\pm$ 0.059	0.0271 $\pm$ 0.0082

4.4 Negative results and ablation summary

We report two classes of negative findings whose details are deferred to Appendix ??, but whose summary is informative.

Initialization variants. We tested 11 alternative initialization strategies, including union-of-amplitude scoring across all train poses (A2–A4), LiDAR-intensity-weighted sampling (A4–A5), strict-AND visibility filtering (B1, B2 with and without test pose), voxel-aware proportional allocation (C1, C1b), and motion-compensated RA-signal modulation (C2). All variants converge within MC noise of the M0 default (cosine-importance + farthest-point sampling on a tightened azimuth FOV cone matching the arcsin-spaced bin grid), with no variant exceeding the $F\pm 1$ inter-frame coherence ceiling. We conclude that *initialization* is not the lever for breaking this ceiling.

Per-pose deformation MLPs. We tested a pose-conditioned per-Gaussian deformation MLP (MLP-A) — input pose 6-DoF + canonical position, output per-point ($\Delta\mathbf{p}, \Delta\alpha$) — at three configurations (raw 6D, strong L2 anchor, NeRF-style positional encoding). All three converge to test CC within ± 0.03 of M0, with positional encoding actually *degrading* mean CC because the higher-dimensional input space encourages train-pose memorization rather than smooth interpolation across the 8 training poses. The conclusion: with only 8 train poses bracketing 1 test pose, an MLP-based deformation field cannot extract additional NVS structure beyond what the static Gaussian point set already captures.

What this implies. Reaching higher novel-view CC will require either (a) higher-rate radar capture (10 Hz+, e.g. K-Radar Paek et al. [2022], RadarML Huang et al. [2024b]) where the inter-frame coherence ceiling is naturally higher, or (b) on-support test splits with denser bracketing of the test pose. We discuss these directions in Sec. ??.

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